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RESEARCH ARTICLE

Integrating media sentiment with traditional economic indicators: a study on PMI, CCI, and employment during COVID-19 period in Poland

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Abstract

Global crises, such as wars or the COVID-19 pandemic, underscore the need for real-time economic monitoring. Traditional economic indicators often fall short, prompting the exploration of alternative data sources, including online and social media content. This study examines the relationship between media sentiment in press articles and traditional economic indicators: the Purchasing Managers' Index (PMI), Consumer Confidence Index (CCI), and average employment in the enterprise sector. We evaluate four pre-trained natural language processing models for sentiment analysis to assess their applicability. The analysis also explores the impact of time shifts in media reporting on the correlation between sentiment scores and economic indicators. Results reveal that a + 24-day shift in article dates produces the strongest correlation with PMI, suggesting media sentiment can predict changes in PMI with a lead time of about 3.5 weeks. Further analysis shows a positive correlation between sentiment scores and the CCI with a + 6-day shift, indicating media sentiment may signal changes in consumer confidence approximately one week in advance. Additionally, a + 70-day shift reveals that media sentiment can predict changes in average employment in the enterprise sector up to 10 weeks before they are officially recorded. These findings highlight the potential of media sentiment as an early indicator of economic trends, emphasizing the importance of considering time dynamics in such analyses. The study demonstrates that sentiment analysis offers valuable insights into economic trends through media reporting, potentially aiding in more timely economic forecasting and decision-making.

Keywords Sentiment analysis · Text processing · Economic forecasting · Purchasing managers' index · Consumer confidence index

Extended author information available on the last page of the article

1 Introduction

The onset of the COVID-19 pandemic precipitated unprecedented disruptions across global supply chains and labor markets. Lockdown measures and mobility restrictions hindered the smooth flow of goods and services, resulting in bottlenecks within production and distribution networks. Simultaneously, widespread job losses and diminished consumer spending exacerbated existing economic challenges. Prior research has examined the varying impacts of different components of supply chain logistics on economic growth, such as overall logistics performance and the efficacy of input and output dimensions [11]. Additionally, studies have encompassed a broader array of variables including each country's stock index, the Purchasing Managers' Index (PMI), and various COVID-19-related factors such as the duration of lockdowns, restrictions on internal and international movement, fiscal measures, and confirmed cases [3].

Certain studies have explored alternative past and prospective trajectories of the pandemic using global statistics on cases and fatalities, alongside analyses of public health and economic policy responses. These analyses have evaluated the economic ramifications of COVID-19 under different scenarios, considering impacts on labor and sectoral productivity, consumption patterns, government expenditure, and risk premia at both national and sectoral levels [21]. Amidst volatile market conditions and evolving consumer behaviors, discerning robust correlations becomes imperative for effective decision-making and policy formulation.

In the pursuit of streamlined data integration and comprehensive insights, the PMI emerges as a pivotal indicator for presenting the overall economic landscape [16]. Widely acknowledged for its ability to gauge the health and trajectory of manufacturing and service sectors, the PMI surveys purchasing managers' sentiments regarding key economic variables such as production, new orders, employment levels, supplier deliveries, and inventory levels. Its significance lies in its timeliness and capacity to capture shifts in economic activity, thereby facilitating informed decision-making for policymakers, investors, and businesses. Studies have validated the utility of PMI in forecasting economic conditions amidst the COVID-19 pandemic [22].

In conjunction with the PMI and employment data, the Consumer Confidence Index (CCI) serves as a vital economic indicator that reflects the overall sentiment and expectations of consumers regarding the state of the economy [27]. The CCI measures the degree of optimism or pessimism that consumers feel about their financial situation and the broader economic environment, including factors such as future income, employment prospects, and general economic conditions. During periods of economic instability, such as the COVID-19 pandemic, shifts in consumer confidence can significantly influence spending behaviors and, consequently, economic growth [6]. The CCI's sensitivity to economic fluctuations makes it a crucial component for analyzing market dynamics and forecasting economic trends [10].

In addition to sentiment analysis through the PMI, employment data serves as a reliable metric for gauging the labor market's response to disruptions. Notably resilient to minor fluctuations, employment data reflects significant market disruptions, such as those induced by the prolonged impact of the COVID-19 pandemic [2, 24], particularly affecting vulnerable segments of society [4, 7]. Consequently, employment data has been utilized both to monitor the pandemic's economic impact and to identify phases of market recovery [15]. By examining the interplay between policy responses and economic indicators such as the PMI and employment data, it becomes possible to uncover underlying correlations that shape economic trajectories, shedding light on the efficacy of various policy measures.

While internal processes largely determine market dynamics for enterprises, external factors, such as media sentiment, can also exert influence. Media sentiment, as portrayed in news articles and social media posts, significantly influences market perceptions and investor sentiment, thereby impacting market volatility, asset prices, and investment decisions [25, 29]. Moreover, media coverage often highlights early-stage processes and phenomena that may culminate in complex cause-and-effect relationships, albeit with a delay in their manifestation [19, 20, 28]. Despite the intricacies of these relationships, exploring correlations between media content and economic conditions remains pertinent. The burgeoning field of market volatility measurement in relation to media sentiment underscores its relevance in economic research, particularly in analyzing phenomena such as the COVID-19 pandemic [1, 9, 13]. Sentiment analysis, a subset of natural language processing, has emerged as a valuable tool for quantifying and analyzing textual data to discern underlying sentiment and trends, enhancing predictive modeling and decision-making processes in economic research [8, 14, 23].

Given the nascent nature of sentiment analysis studies, there remains a gap in evidence concerning the relationship between media sentiment and economic conditions across different countries. Therefore, this study aims to investigate the correlation between national media sentiment, PMI, CCI and employment data in Poland, spanning the period from January 2019 to December 2021. The significance of this study lies in the increasing demand for timely and accurate economic predictions amidst economic uncertainty and volatility [17]. Understanding the interplay between media sentiment and traditional economic indicators can provide valuable predictive insights into economic trends and market dynamics, thereby informing policy-making, investment decisions, and risk management strategies.

2 Data and methods

2.1 Data and preprocessing

This section provides a comprehensive overview of the data collection method, specifically focusing on the origins of news articles and economic data. The text provides an explanation of the preprocessing procedures employed in sentiment analysis, as well as the analytical approaches utilized to establish a correlation between media sentiment and economic data.

2.2 Articles and sentiment assessment

The data used for our analysis originates from newspaper articles that were published in Poland throughout the period from January 1, 2019, to December 31, 2021. Analyzed were a total of 190 369 articles (Fig. 1) published in various regions of the country. The time interval used for analyzing the published texts is one day. The data we retrieved included the publication date of the article, its main title, a subtitle providing a brief summary, and the full text description of the article.

Sentiment analysis was performed using four pre-trained models specifically designed for Polish language processing. These models were selected based on their complementary strengths and domain expertise, allowing us to capture different aspects of sentiment expression in economic news. These models were chosen for their adaptation to different thematic contexts, including financial, political and general social. The selected models had already been rigorously evaluated and fine-tuned by their developers on large datasets relevant to their thematic domains (e.g., financial news, social media, political texts). In this study, the models were applied in their original form to assess their potential in predicting changes in economic indicators in Poland. Additional fine-tuning or validation was not possible due to the limitations of the available historical data, which did not cover a wide enough temporal range, including pre-pandemic periods and periods of economic stability.

The first model was Finance Sentiment PL by bards.ai (s1) which is a financial sentiment model that provides precise sentiment ratings that indicate categorization confidence. This model is effective for financial news sentiment analysis. It

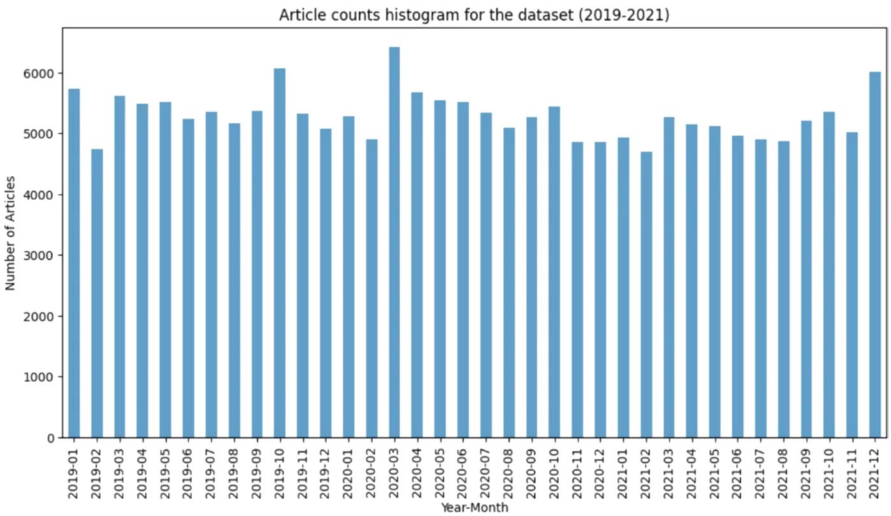


Fig. 1 Monthly distribution of article counts from January 2019 to December 2021, showing variability in the number of articles published each month, with notable peaks in April 2020 and December 2021, reflecting heightened media activity during these periods

was trained on an extensive corpus of financial news articles and reports. This model excels in detecting subtle sentiment variations in economic contexts and provides detailed confidence scores alongside sentiment ratings.

The second one was Twitter sentiment PL (s2), designed for sentiment analysis for Polish social media communications, notably Twitter. This model was developed using over one million Polish tweets, making it particularly adept at processing informal language patterns and capturing rapid sentiment shifts. Its training on social media content enables it to effectively interpret contemporary language usage and emerging linguistic patterns in news articles.

The third model was A Herbert language model-based sentiment analysis model for Polish texts (s3), adapted to different text contexts, from formal documents to more liberal written forms. This model's strength lies in its adaptability to various writing styles and contexts, making it particularly valuable for analyzing news articles that span different topics and writing formats.

The last one was PaReS-sentimenTw-political-PL by eevvgg (s4), which was a model for Polish political text sentiment analysis. It specializes in political discourse analysis, offering insights into the political aspects of economic news. Its training on political texts enables it to capture subtle nuances in policy-related discussions that might impact economic sentiment.

Because of the different outputs of the models and for the purpose of achieving consistency and comparability between models, we scaled sentiment ratings to -1 for negative, 0 for neutral, and 1 for positive. It allowed for consistent sentiment data analysis and explicit model output comparisons.

2.3 Economic data

The Purchasing Managers' Index (PMI), Customer Confidence Index (CCI) and employment data were collected from the General Statistical Office in Poland. Figure 2 shows the development of three normalized time series: the PMI, the CCI and employment data from January 2019 to the end of December 2021. The indicators have been normalized to values from 0 to 1 for comparison purposes.

All three indicators show sharp declines around March and April 2020, which can be linked to the global impact of the COVID-19 pandemic on the economy. Prior to early 2020, all indicators remain relatively stable with slight fluctuations. In contrast, the pace of recovery from the drastic decline differs. The PMI recovers quickly and maintains a relatively stable level with slight fluctuations until the end of the period. Employment recovery is slower compared to the PMI and shows greater fluctuations, suggesting that the labor market may have experienced greater difficulty in returning to its pre-pandemic state. The recovery of the CCI is the most volatile of the indicators presented, which may indicate volatile consumer sentiment and uncertainty about the economic future.

Recovery patterns are showing volatility, as seen by the rebound and subsequent movements seen in the PMI and employment rate. However, the CCI is still below baseline levels, indicating a continuing state of consumer anxiety.

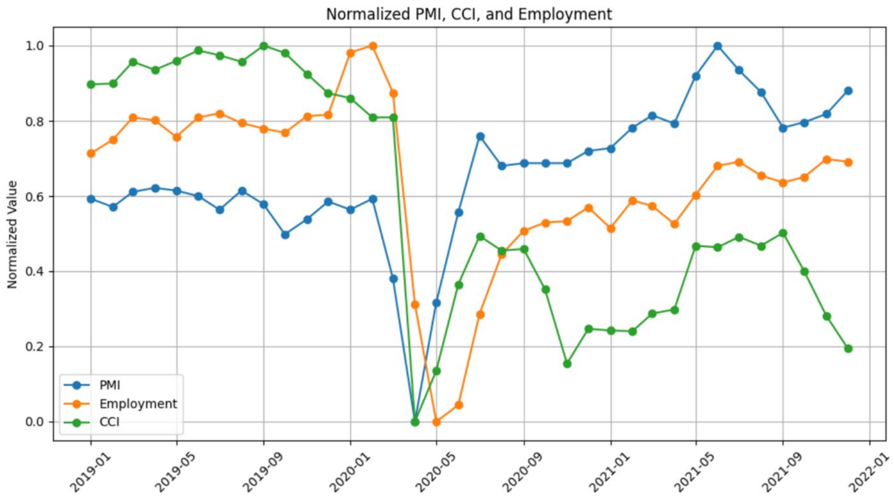


Fig. 2 Normalized trends of the Purchasing Managers' Index (PMI), Consumer Confidence Index (CCI), and Employment in the enterprise sector from January 2019 to December 2021, showing the significant drop across all indicators during the onset of the COVID-19 pandemic in early 2020, followed by varying recovery patterns

2.4 Correlation (with shift)

In this study, we calculate the Pearson correlation between the sentiment of news articles, the PMI, employment and CCI using temporal shifts. Analyzing correlations with temporal shifts allows for investigating the relationship between media sentiment and economic indicators over different time frames. It enables the identification of both leading and lagging impacts, offering valuable insights into the relationship between economic trends and media sentiment dynamics. Temporal shifts refer to the process of modifying the dates of articles in relation to the dates of the analyzed economic data. The shifts might be either negative or positive, indicating whether the articles are compared with earlier or later indicators, respectively.

Negative shifts involve moving the dates of news articles backward in time relative to the dates of the economic indicators. For instance, a shift of − 10 days means that news articles dated March 20 are compared with economic data from March 10. This method allows us to investigate whether changes in the PMI, CCI and employment indicators precede changes in media sentiment.

The strong correlation observed with negative shifts suggests that the media are reacting to previous economic conditions as reflected by economic indicators. For example, if a shift of − 10 days shows a high correlation, it can be inferred that today's changes in PMI, CCI and employment can predict the sentiment of newspaper articles in 10 days. This indicates a lag effect, where economic indicators lead to changes in media sentiment.

For instance, we consider a scenario where a significant drop in the PMI is observed. If, after a shift of 10 days, the sentiment of news articles becomes more negative, it implies that the media is reacting to economic conditions that were present 10 days earlier. This finding can be important in understanding how quickly economic news impacts media reporting and public perception.

Positive shifts include moving the dates of news articles forward in time relative to the dates of the indicators. For example, a shift of + 10 days means that news articles from March 10 are compared with PMI, CCI and employment data from March 20. This approach helps us examine whether changes in media sentiment can precede changes in the PMI and employment indicators.

This is most relevant in our context, as the strong correlation observed with positive shifts indicates that sentiment in newspaper articles has the potential to predict future changes in indicators. For example, if a positive shift by 10 days results in a high correlation, it can be inferred that the sentiment of newspaper articles from 10 days ago can be used to predict today's PMI, CCI and employment values. This indicates an anticipatory effect, in which media sentiment can serve as an early indicator of upcoming economic trends. For example, if the sentiment of newspaper articles becomes more positive and a corresponding increase in indicators is observed 10 days later, this means that media sentiment may be picking up early signs of economic improvement.

3 Results

This section examines the correlation between media sentiment, as derived from Polish news articles, and key economic indicators—PMI, CCI, and employment data—over the period from January 2019 to December 2021. Utilizing four distinct pre-trained natural language processing models specifically designed for sentiment analysis, we assess the extent to which variations in media sentiment can act as early indicators of shifts in these economic metrics. The findings underscore the varying strengths of correlation across different time shifts, providing valuable insights into the predictive capabilities of media sentiment for forecasting economic trends.

Each economic indicator is explored in detail, highlighting the temporal dynamics that define the relationship between public sentiment, as reflected in media coverage, and formal economic metrics. Articles were assigned sentiment values of 1 for positive, 0 for neutral, and -1 for negative by each model. An overall sentiment score (SAr) was also computed as the mean sentiment across all models to ensure consistency and comparability.

Analysis of the number of articles (Fig. 3) reveals notable differences in the distribution of positive, negative, and neutral classifications across the models. These variations reflect the distinct design and contextual focus of each sentiment analysis model, emphasizing their unique contributions to understanding media sentiment trends.

The S4 model identified the highest number of articles as negative, while the S3 model classified more articles as positive compared to the others. In contrast, the S1 and S2 models predominantly categorized articles as neutral. These differences align with the specific adaptations of each model to distinct types of text and contextual nuances, leading to varying distributions of sentiment classifications.

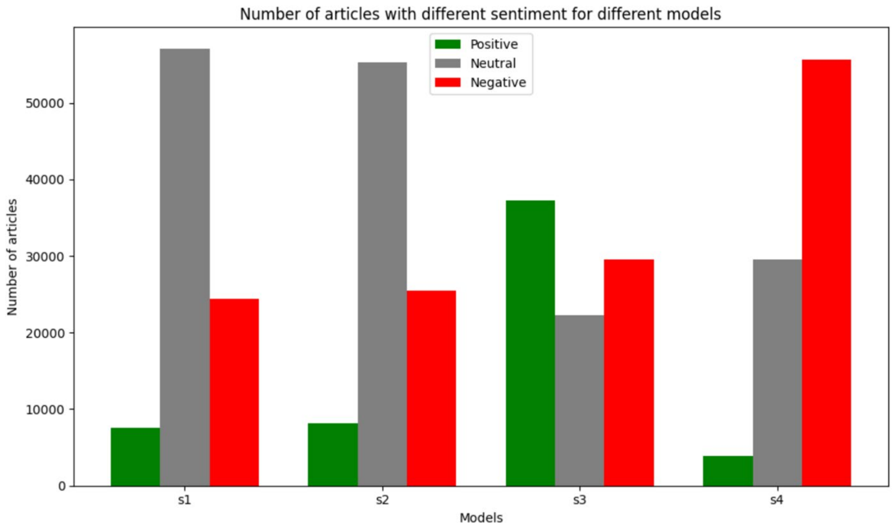


Fig. 3 Comparison of the number of articles classified as positive, neutral, or negative across four sentiment analysis models (S1, S2, S3, S4). The models exhibit varying distributions of sentiment classification, with models S1 and S2 showing a predominance of neutral sentiment, while S3 and S4 display a higher proportion of negative sentiment, particularly in model S4

Figure 4 provides an overview of the evolution of sentiment in media articles over time as analyzed by the different sentiment models. The figure displays the aggregated monthly average number of articles for all models. Analyzing these trends reveals emerging patterns and characteristics across sentiment categories. Notably, periods of significant fluctuation in sentiment are evident. For instance, the first quarter of 2020—marked by the onset of the COVID-19 pandemic in Poland—saw a sharp increase in the number of articles across all sentiment categories. This surge likely reflects the widespread implementation of restrictions, heightened social concerns, and economic uncertainties during this period. Similarly, a noticeable rise in media activity occurred in the summer of 2020, which coincided with a complex political situation in Poland and the subsequent wave of the COVID-19 pandemic.

For the s1 model, neutral articles dominate most of the time (Fig. 4a). This aligns with the expected analysis in a financial context, where the language is expected to be primarily objective. Periodically, there is an observable fluctuation in the number of positive and negative articles, which may indicate periods of heightened market instability or significant economic events that generate higher responses.

Sentiment distribution for articles classified using the s2 model indicates a relatively even distribution between all three categories of sentiment, which may suggest the ability to respond quickly to changing social media moods (Fig. 4b). There are several peaks in the positive and negative classifications that presumably correlate with important social or political events.

The s3 model shows significant variation in the positive and negative classifications, with clear peaks that can reflect cultural or political events that have a

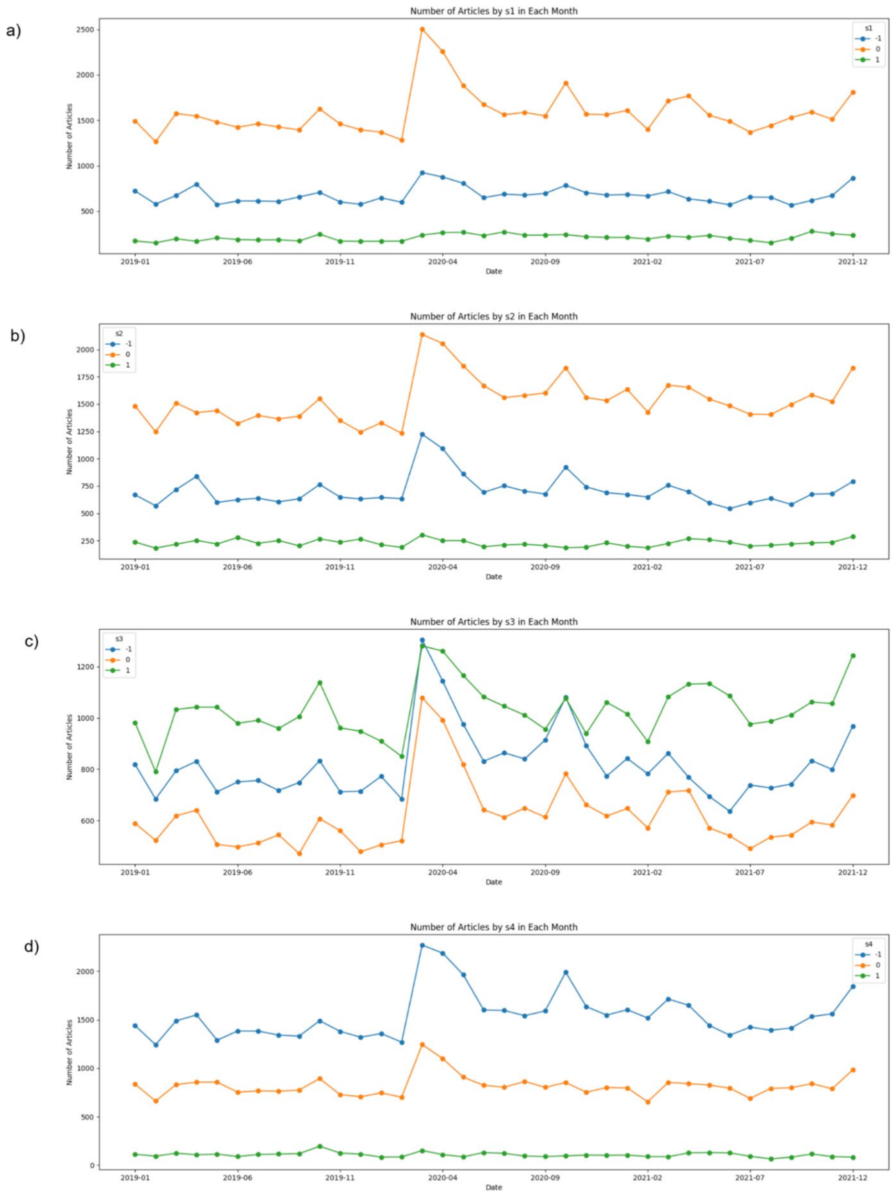


Fig. 4 Monthly trends in the number of articles categorized by sentiment scores (−1, 0, 1) for models **a** s1, **b** s2, **c** s3, and **d** s4, from January 2019 to December 2021. Each plot highlights distinct patterns in sentiment distributions over time, with notable fluctuations, including peaks and troughs that may correspond to significant external events

notorious impact on public opinion (Fig. 4c). The s4 model is characterized by a large number of negative articles, which is expected for a model focused on political texts, where criticism and conflict are more frequent (Fig. 4d). These few articles that are both positive and neutral highlight the specificity of political discourse, which frequently centers on negative emotions.

By examining the quantity of articles exhibiting positive and negative emotion throughout the progression of the epidemic (Fig. 5), the data revealed the trajectory of the pandemic. We have observed an increase in negative articles in March 2020, which coincides with the confirmation of the first case of COVID-19 in Poland. A further marked increase in the number of negative articles was recorded around November 2020, which may reflect the second wave of pandemics, characterized by a significant increase in cases and the introduction of additional restrictions.

For most of 2021, especially from summer to autumn, the number of negative articles stabilized with smaller increases and decreases, which may indicate further waves of infections, while also increasing public adaptation to life in pandemic conditions, as well as the visible impact of vaccinations. The distribution of articles on the chart shows how the media responded to the pandemic, reflecting both moments of worsening health and more optimistic periods of managing the crisis. These fluctuations correlate with key pandemic moments that have had a direct impact on general social and media sentiment.

3.1 Correlation with purchasing managers' index (PMI) (including shifts)

In the correlation analysis with the PMI, sentiment analysis outcomes were compared with PMI data, a key indicator of economic trends. This analysis aimed to uncover potential relationships between the sentiments expressed in articles and the economic conditions represented by the PMI. By employing correlation coefficients and accounting for time lags, the study sought to identify the strongest connections between article sentiment (analyzed using different language models) and PMI values.

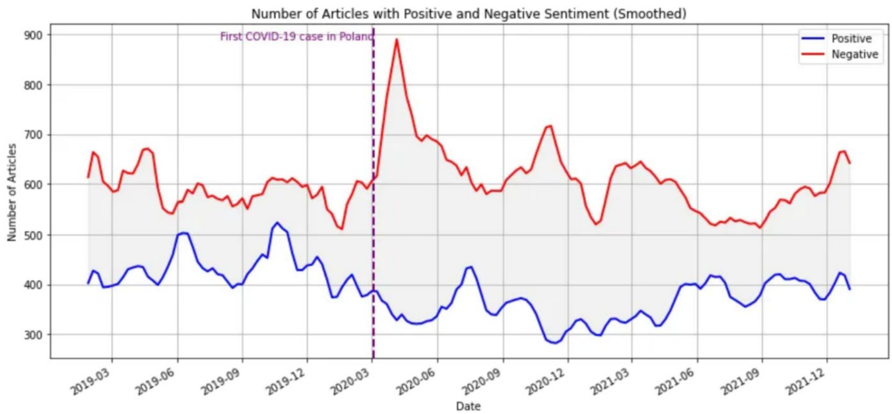


Fig. 5 Smoothed trends of articles with positive and negative sentiment from March 2019 to December 2021, highlighting the impact of the first confirmed COVID-19 case in Poland (marked by the purple dashed line in March 2020). The data shows a sharp increase in negative sentiment following the outbreak, peaking in mid-2020, while positive sentiment articles decreased and fluctuated at lower levels during the same period

Figure 6a illustrates the normalized PMI index alongside average sentiment scores (SAr) over time. The PMI index exhibited moderate fluctuations before showing a slight decline prior to 2020. In mid-2020, the PMI experienced a sharp drop due to the COVID-19 pandemic, which severely disrupted global supply chains and economic activities. Following this significant decrease, the PMI rebounded swiftly, reaching a peak in mid-2020 and maintaining stability through the end of the year. By 2021, the PMI's stabilization suggests a recovery in procurement and sales activities, reflecting the resilience of businesses in adapting to post-pandemic challenges.

By showing average article sentiment on the same chart, it exposes more volatility than PMI. Negative mood dominates articles throughout the period since sentiment levels are negative. As with the PMI, sentiment fluctuates with the start of the pandemic, declining significantly in the first half of 2020, which may reflect the outbreak's depressing mood. Despite major changes, sentiment is rising in 2021.

The results of the PMI correlation with the sentiment of articles indicated a positive relationship between them. Specifically, the s2 model shows the highest correlation with the PMI index among all sentiment analysis models. For the rest of the models, the correlation rates are lower.

Figure 6b shows the correlation values between the mood in articles and the PMI index based on the number of days of shift for four different sentiment analysis models. The analysis of the results shows that different models present different levels of correlation depending on the time shift.

Model s1 shows relatively low correlation values, which begin to rise in positive shifts, peaking around + 74 days of a shift with a maximum value of about 0.4717. The s2 shows the highest correlation values of all models, with a marked increase in correlations of about − 100 days and peaking in the range of 0 to + 25 days of shift, reaching a maximum of about 0.6177 with a shift of 24 days (*p* value:0.000060) which is the most significant result of all models and shifts. Model s3 shows higher correlations than s1, but lower than s2, reaching a maximum of about 0.5219 with a 24 day shift. In turn, the model s4 presents lower correlation values compared to the models s2 and s3. Correlations for s4 are generally negative for shifts in the range of − 100 to 0 days, with a maximum correlation value of approximately 0.0487. The only model showing a predominantly negative correlation, with the lowest value at the maximum positive shift.

The findings indicate that the s2 and s3 models perform well in forecasting PMI changes based on article sentiment, particularly within the range of 0 to + 25 days of shift. These models indicate that the sentiment expressed in articles can serve as an early indicator of future changes in PMI, approximately 25 days in advance. The s4 model, with lower correlation values, shows less efficiency in the context of predicting changes in PMI.

General increase in correlation was observed from − 100 days of shift to about + 24 days shift, followed by a decrease. The highest correlation value reaches 0.6177 at + 24 days shift, indicating the strongest relationship between article sentiment and PMI at the moment.Other high correlations for s2 include shifts of 10 days (0.617152, *p* value: 0.000061) and 22 days (0.614487, *p* value: 0.000067). This result suggests that changes in article sentiment most effectively predict changes in PMI with a delay of 24 days. In other words, the sentiment expressed in the articles

Fig. 6 Temporal dynamics between the Purchasing Managers' Index (PMI) and average article sentiment are shown, illustrating their alignment and variability over time (**a**). The shift-dependent correlation analysis for four sentiment models reveals how the relationship with PMI changes across time lags, identifying periods of stronger or weaker correlation (**b**). The peak correlation shift and its statistical significance are further ▶

can serve as an early indicator of future economic conditions measured by the PMI, approximately 24 days ahead.

The p-test values remain small in the range of −30 to+50 days, which supports our findings and reflects a good correlation between the sentiment in the articles and the PMI index during this period (Fig. 6c). After exceeding this interval (from±50 days), the correlation is significantly reduced, suggesting that the relationship between article sentiment and PMI is most significant in the short term.

3.2 Correlation with customer confidence index (including shifts)

As observable in Fig. 7a, the consumer confidence index initially shows slight fluctuations, remaining at a relatively high level until mid-2019. Then we see a marked decline in CCI values in the middle of 2020, which coincides with the onset of the COVID-19 pandemic. After this significant decline, the CCI has somewhat rebounded, but remains lower than before the pandemic, showing greater volatility in subsequent periods.

By showing on the same chart the average sentiment of articles, we see that this sentiment also shows greater fluctuations than the CCI. As with the PMI, significant declines and rebounds in the CCI are associated with marked changes in the sense of articles. The observed fluctuations in sentiment may be an indicator of changing public mood in response to changes in the state of the economy, especially in the context of the COVID-19 pandemic.

The results of the CCI correlation with the sentiment of articles indicate a positive relationship between them. The analysis showed that at a time when consumer confidence is falling, the sentiment expressed in articles also becomes more negative. Especially during the period of initial pandemic disruptions, both variables show synchronized declines.

Figure 7b shows the correlation values between sentiment in articles and the consumer confidence index (CCI) depending on the number of days of shift for four different sentiment models.

The s1 model shows only negative correlations, with peak around +52 days (−0.4581, *p* value: 0.0056). The s2 model shows a marked increase in positive correlation from about +24 days of shift, reaching peak values around shift +30 days (0.3778, *p* value: 0.0230), and then reaching a relatively stable level of about 0.3. The s3 shows a similar trend to the s2, but with slightly higher correlation values. The correlation grows significantly from about +20 days of the shift, reaching a maximum of 0.3987, *p* value: 0.0194 in 76 days and then remaining at this level for most of the period of analysis.

The s4 features the highest correlation of all models. The correlation grows gradually from about −100 days of shift, peaking at about 0.7004 at a shift of about 0,

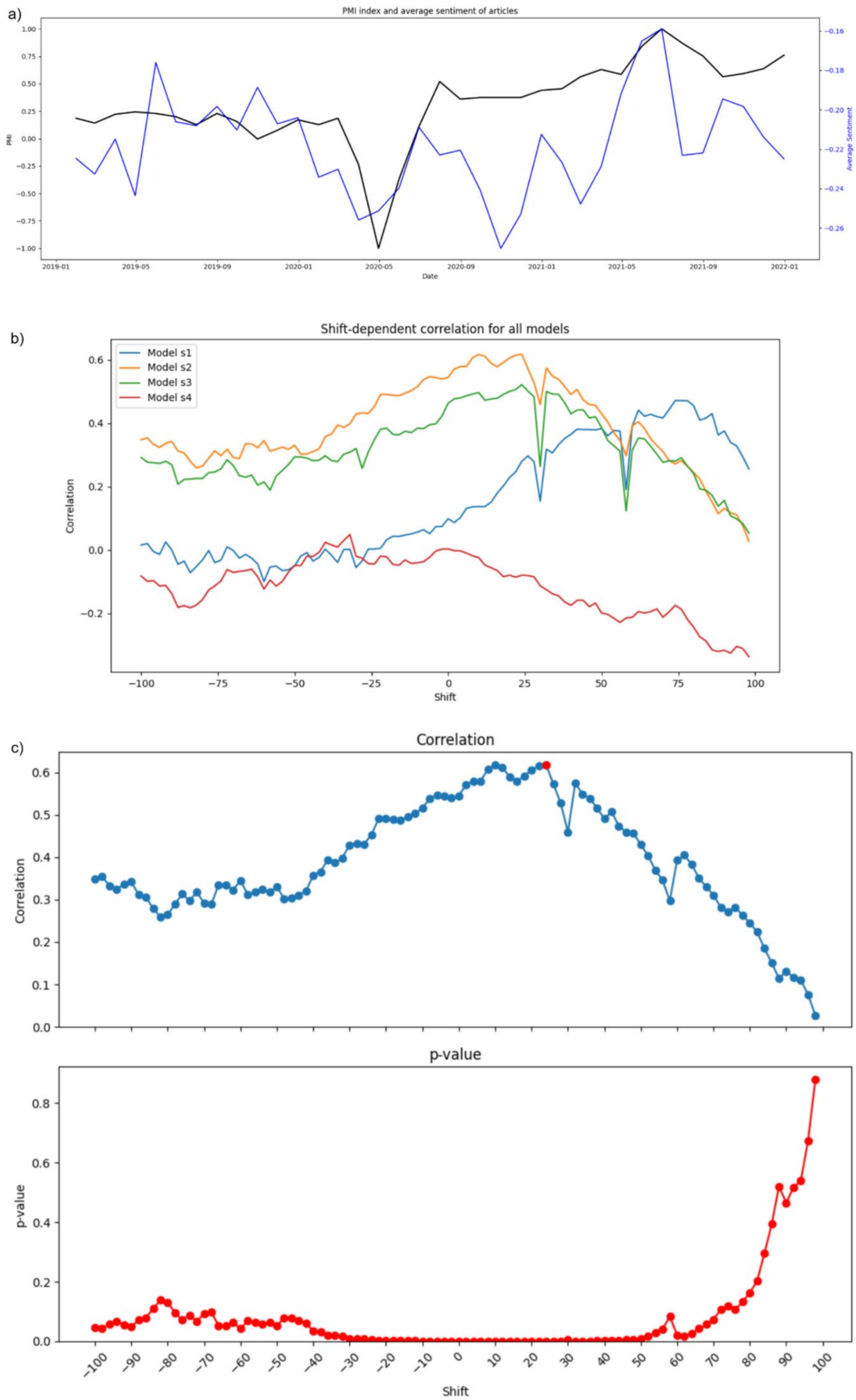

a)
PMI index and average sentiment of articles
PMI
Average Sentiment
Date
b)
Shift-dependent correlation for all models
Model s1
Model s2
Model s3
Model s4
Correlation
Shift
c)
Correlation
Correlation
p-value
p-value
Shift

Fig. 7 The relationship between the Customer Confidence Indicator (CCI) and the average sentiment of articles is depicted, showing their temporal co-variation over the analyzed period (**a**). The shift-dependent correlation analysis demonstrates how the association between the CCI and sentiment models (S1, S2, S3, S4) evolves across different time lags (**b**). The maximum correlation shift is further analyzed alongside its statistical significance, providing insights into the temporal offset where the relationship is strongest (**c**)

and then remaining high for the remainder of the period of analysis. High correlation values for the s4 model suggest that it is most effective in predicting changes in the CCI based on the sentiment expressed in the articles.

The results indicate varying levels of predictability among models in forecasting CCI changes, with the s4 model demonstrating the highest level of predictability. Models s2 and s3 also show moderate efficiency, while s1 shows the lowest predictability. The variability underscores the critical importance of choosing the appropriate model based on data characteristics and analytical objectives.

Figure 7c shows a more detailed correlation course for model s4 depending on the number of days of shift. A general increase in correlation was observed from −100 days of shift to approximately +25 days shift, followed by a slight decrease and then stabilization. The highest correlation value reaches 0.7191, p value < 0.001 with a shift of +6 days, indicating the strongest relationship between article sentiment and CCI at the moment. This result suggests that changes in the sentence of articles most effectively predict changes in CCI with a delay of 6 days. In other words, the sentiment expressed in the articles can serve as an early indicator of future economic conditions measured by the CCI, approximately 6 days ahead.

Like the PMI findings, which showed a strong correlation of 0.6177 with a 24-day shift, the CCI results also suggest that media mood can serve as a predictor for future changes in economic data. These findings indicate that changes in media attitude could possibly serve as early indicators of economic circumstances, but the timing and strength of these indicators may vary across different measures. The p-tests remain small in the range of −50 to +50 days, which supports our findings and reflects a good correlation between the sentiment in the articles and the CCI during this period.

Once the time period of ±50 days is surpassed, the correlation between the mood of articles and the CCI decreases. This indicates that the relationship between the tone of articles and the CCI is most notable in the near term. The findings for PMI also align with this pattern, since the association was best in the short term.

3.3 Correlation with labor market—average employment in the enterprise sector (including shifts)

The Fig. 8a illustrates a comparison between the average employment and the average sentiment of articles over the analyzed period. By the start of 2020, employment, which serves as a crucial measure of the labor market, exhibited stability. However, it experienced an important decline because of the COVID-19 pandemic. During the later time, the employment rate is seeing a gradual upward trend, but there are occasional changes.

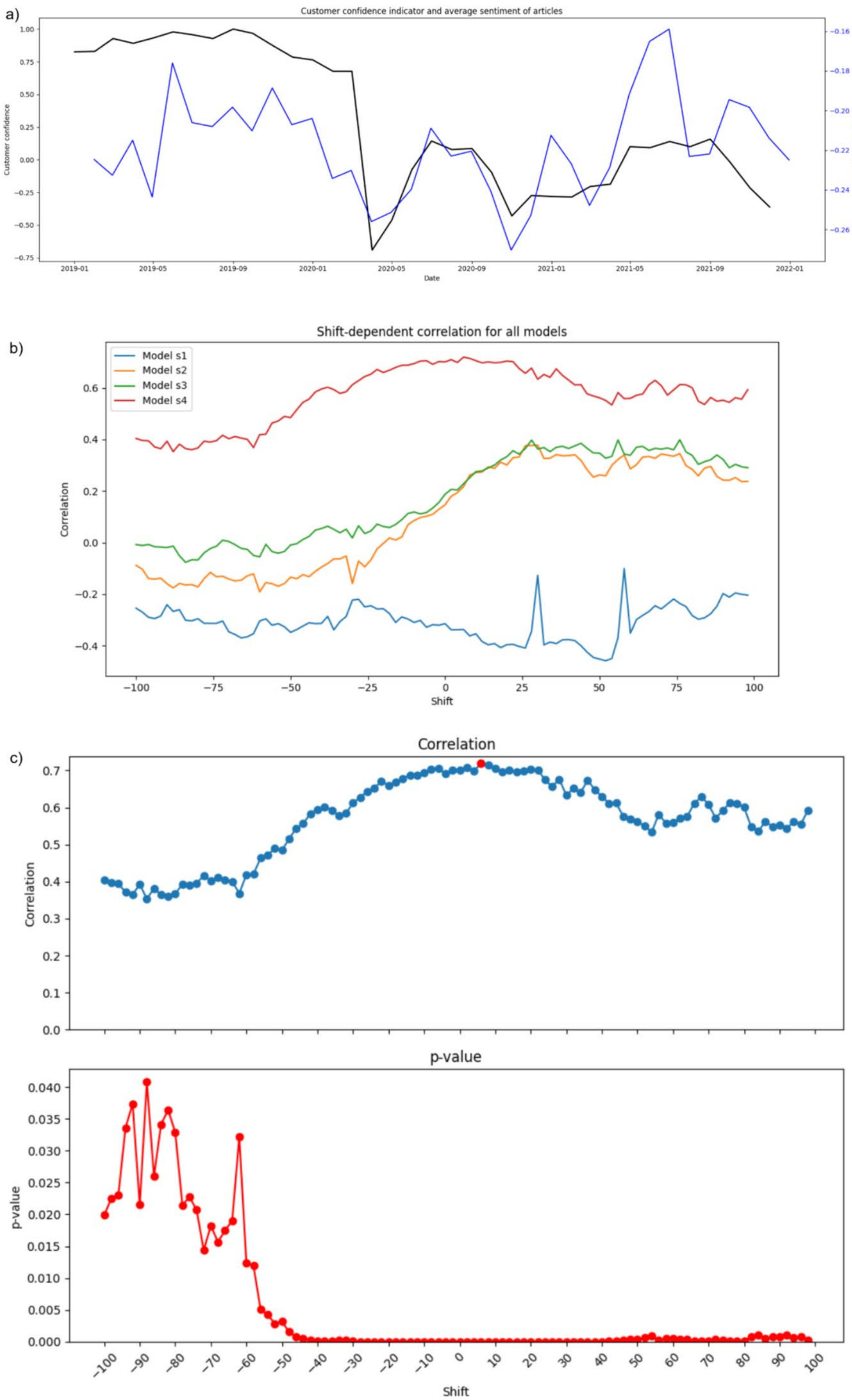

a)
Customer confidence indicator and average sentiment of articles
Customer confidence
Average Sentiment
Date
b)
Shift-dependent correlation for all models
Model s1
Model s2
Model s3
Model s4
Correlation
Shift
c)
Correlation
Correlation
p-value
p-value
Shift

Fig. 8 The connection between employment levels and average article sentiment over time highlighting trends and fluctuations in their relationship (**a**). The shift-dependent correlation analysis investigates how the association with employment varies across different time lags for the sentiment models (S1, S2, S3, S4), revealing patterns of temporal dependency (**b**). The strongest correlation and its statistical significance are identified, pinpointing the optimal time lag for the relationship (**c**)

As with the PMI and CCI, significant changes in the employment rate are correlated with clear changes in article sentiment. These fluctuations in sentiment may reflect changing social mood in response to changes in the economy, especially in the context of pandemics.

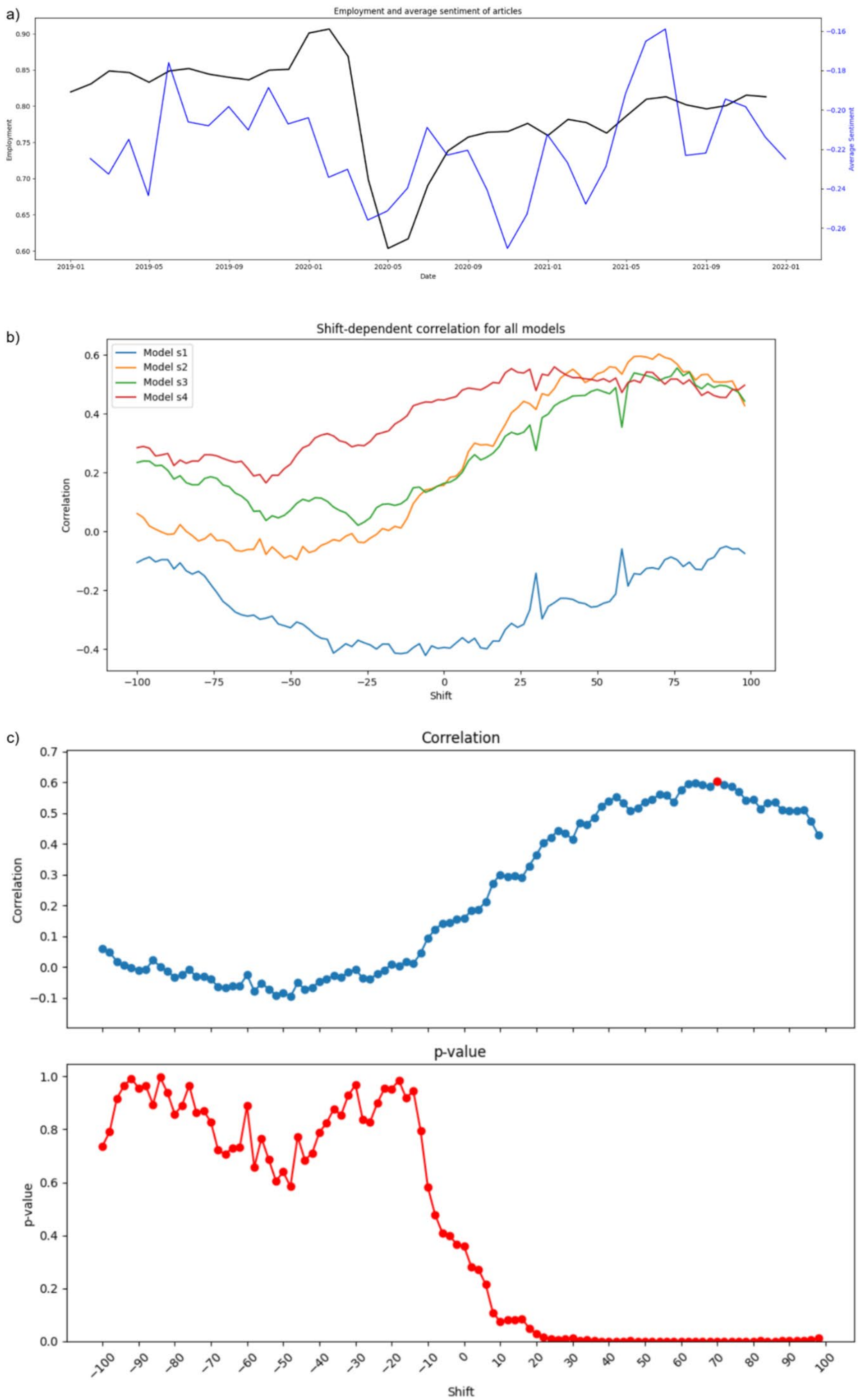
The s1 model shows the lowest correlation values, which initially remain at around −0.4. The correlation drops to its lowest level of about −0.4211 at a shift of −6 days. In all periods the correlations are negative. The s2 model presents the highest positive correlations of all models. It shows a marked increase in correlation from about +50 days of shift, reaching peak values around the shift of +70 days (0.6033, *p* value: 0.0001). The s3 shows a similar trend to the s2. The correlation grows from about +20 days of shift, reaching a maximum of about 0.5559, *p* value: 0.0006 in the area of +76 days shift and then remains at this level for most of the period of analysis. In s4 the correlation grows gradually peaking at about 0.5597, *p* value: 0.0004 at a shift of +36 days (Fig. 8b).

Figure 8c shows a detailed correlation between articles sentiment and employment rate in enterprises depending on the number of shift days for model s2. We may see the overall increase of the correlation in the analyzed period day shift to about +70 days, where we can notice the highest correlation value (0.6033, *p* value: 0.0001), suggesting the strongest relationship between article sentiment and employment rate at this shift. This indicates that changes in the meaning of articles can effectively predict changes in employment about 70 days in advance..The p values are low for the range of shifts from −20 to+100 days, which means that the correlation results in this range are statistically significant.

4 Discussion

The results demonstrate statistically significant correlations between PMI, employment rate, and daily press article sentiment. These relationships vary for PMI, employment index, and CCI, but they all stem from scientific discourse and study. Nevertheless, the feelings expressed in the media provide valuable insights into the economic landscape.

While our study utilized pre-trained models without additional fine-tuning, the validation of their effectiveness emerged through the analysis of correlations with actual economic indicators. The best correlation results for the PMI and employment data were achieved with the s2 model (bardsai/twitter-sentiment-pl-fast). The strongest correlation of 0.6177 for PMI was a 24-day shift, demonstrating that social media mood is the best indicator of PMI changes 3.5 weeks in advance. For employment, it was 70 days, suggesting sentiment changes precede employment changes by 10 weeks. The s4 model (eevvgg/PaReS-sentimenTw-political-PL), specifically


a)
Employment and average sentiment of articles
Employment
Average Sentiment
Date
b)
Shift-dependent correlation for all models
Model s1
Model s2
Model s3
Model s4
Correlation
Shift
c)
Correlation
Correlation
p-value
p-value
Shift

designed for political literature, yielded the highest correlation values in the particular case of CCI. The correlation coefficient reached its maximum value of 0.71 when a time shift of 6 days was used. These findings indicate that the emotional tone of articles could serve as an early predictor of forthcoming changes in consumer confidence, with a brief lead time of approximately one week.

Understanding these different temporal dynamics is crucial for interpreting the predictive power of media sentiment. This is consistent with the notion that by looking at article sentiment, we can sometimes see early signs of changes in the economy, even before formal economic indicators record these changes. The results for employment show the longest lag time compared to the PMI and CCI, suggesting different dynamics in the impact of article sentiment on these economic indicators. This variation in lag time can be attributed to the nature of the indicators themselves.

As a forward-looking index, the PMI is likely to be more sensitive to immediate changes in sentiment. If media sentiment becomes more upbeat, this may indicate that purchasing managers are also feeling an improvement in the economy, which is later reflected in the PMI.

Consumer confidence can react quickly to news events, reflecting short-term changes in public perceptions. If media sentiment is positive, consumers may feel more confident about the future, which quickly affects their confidence. On the other hand, changes in hiring often result from long-term trends and company decisions. If media sentiment turns more negative, it may indicate impending economic difficulties that companies will begin to feel and respond to, such as through redundancies, but this process takes longer.

These results are also consistent with previous studies, which have shown that the analysis of mood in the media can be a useful prognostic tool. For example, research on using Google Trends to predict PMI [12] and analysis of survey headers [18] show that alternative data sources can provide valuable forecasting information.

Previous studies on PMI forecasting, in particular using data from Poland, have had varying degrees of success. The ability to predict the future became better at a time when the economy was less stable [26]. In particular, during global economic turmoil, PMI levels have shown predictability based on Google Trends [12] or analysis focused on data from survey headers [18], representing alternative sources of mood for the media. Our study is consistent with these findings, revealing statistically significant links between media sentiment and PMI values. Considering the usefulness of PMI as a tool to predict GDP in some cases [5], receiving preliminary signals, such as media signals can offer a valuable strategy to stabilize economic conditions in turbulent times.

The results also show how different models of sentiment analysis can have different applications and effectiveness depending on the specific economic context and the type of data analyzed. The use of the s2 model, designed for rapid analysis of social media texts, is useful in the context of PMI and employment, which may be due to the dynamic nature of these indicators and their sensitivity to rapid changes in the mood expressed in social media. In turn, the s4 model, which specializes in analyzing the sentiment of political texts, shows its value in the context of CCI, which may be linked to the more stable and less dynamic nature of the consumer confidence index, which reflects long-term consumer sentiments and prospects.

The use of different models of sentiment analysis revealed that different models show different effectiveness in predicting changes in economic indicators. The common feature of these models is that they have been trained to recognize and analyze mood in Polish, however, each of them has different specializations and is adapted to different types of texts and contexts, which makes them tools especially useful in specific situations. Thanks to this diversity, they can be effectively used to study relationships between media and economic indicators depending on the context of the analysis. Each of these models is adapted to specific types of text and contexts, resulting in different distribution of sentiments in the classification. Therefore, interpreting the results of each model requires an understanding of both the mechanisms of the model and the nature of the data on which it was applied.

The methodological approach employed in this study, utilizing multiple pre-trained sentiment analysis models, offers several advantages. The diversity of the models' specializations allowed us to capture different aspects of economic sentiment, while their complementary nature helped mitigate individual model biases. This multi-model approach, combined with the temporal analysis, provides a foundation for future research in sentiment-based economic forecasting, particularly in markets where extensive historical data may not be available for model fine-tuning.

However, future research directions should address some limitations of the current study. First, while our focus on news articles provided reliable and structured data, incorporating alternative media sources such as social media platforms and specialized economic blogs could offer additional perspectives and potentially improve prediction accuracy. Second, extending the analysis to include pre-pandemic periods and times of economic stability would provide a more comprehensive understanding of the models' predictive capabilities across different economic cycles. Additionally, developing more sophisticated statistical frameworks that account for autocorrelation and external economic factors could further validate the causal relationships suggested by our time-lag analysis. Such extensions would contribute to building a more robust methodology for sentiment-based economic forecasting.

The relationship between media sentiment and economic indicators is complex and potentially influenced by various external factors. While our analysis focuses on direct correlations and time-lag relationships, we acknowledge the presence of potential autocorrelation in the time series data. The varying lag times observed for different indicators (24 days for PMI, 6 days for CCI, and 70 days for employment) suggest distinct underlying mechanisms in how media sentiment interacts with different aspects of the economy. These differences might reflect the inherent nature of each indicator: PMI being more responsive to immediate market sentiment, CCI reflecting rapid changes in public perception, and employment showing more inertia due to institutional factors.

A limitation of this study is the relatively short analysis period, which predominantly focuses on the pandemic. To comprehensively assess whether media sentiment serves as a reliable predictor, it is crucial to obtain a more extensive dataset encompassing periods of both economic crises and stability. Expanding the temporal range of the data would provide a more nuanced understanding of the media's predictive capacity under diverse economic conditions.

In conclusion, the results of this study underline the importance of sentiment analysis as a predictive tool in economics. Various sentiment analysis models can provide valuable insights on future changes in PMI, CCI and employment, and their application can help analysts and policy makers to better understand and respond to changes in the economy.

5 Conclusions

Our analysis of the correlation between press sentiment and key economic indicators—PMI, CCI, and employment rate—yields significant insights that could enhance the prediction of future economic trends and the understanding of market dynamics. The study confirms the hypothesis that news sentiment is closely linked to economic activity, offering a valuable tool for analyzing and forecasting economic trends. By identifying shifts in media sentiment ahead of traditional economic indicators, there is potential for earlier intervention and more proactive economic management.

The application of pre-trained sentiment analysis models as early indicators for changes in PMI, CCI, and employment rate suggests that these models can be instrumental in detecting potential economic shifts more swiftly than traditional methods. This capability is particularly valuable for analysts and policymakers, providing them with a powerful tool to anticipate and respond to changes in key economic metrics.

Importantly, our objective is not to replace existing economic indicators but to complement them by exploring the relationship between media sentiment and official economic indices. This approach aims to enhance the understanding of both current and near-term economic conditions. The study demonstrates that integrating media sentiment as an additional metric can potentially improve the accuracy of economic forecasts and inform more effective decision-making.

By incorporating alternative data sources like media sentiment, analysts and policymakers can better navigate the complexities of economic forecasting, reducing the risks associated with relying solely on traditional data. This complementary approach promises to deepen our understanding of economic patterns and enhance our ability to respond to economic challenges.

Future research should continue to examine these issues, taking into account different contexts and data sources, to better understand the dynamics of how media sentiment affects economic indicators. Additionally, the inclusion of broader datasets covering multiple economic phases, including stability and crises, could greatly enhance the robustness of the findings. Such an approach would allow researchers to assess the long-term reliability and comprehensiveness of media sentiment as an early predictor of economic trends, offering a more comprehensive economic forecasting framework.

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Data availability The data used in this study are not available due to lack of licensing.

Software availability Available on request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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